

Gradient Descent Intuition

youtube_notes

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Abstract

This sample note shows how a short transcript excerpt becomes a structured educational LaTeX note.

1 The Basic Picture

Suppose a model is controlled by parameters θ and we measure its error with a loss function $L(\theta)$. Gradient descent updates the parameters by repeatedly taking a step that tries to reduce the loss.

Intuition

The gradient $\nabla L(\theta)$ points in the direction of steepest local increase. To move downhill, use the opposite direction.

2 Update Rule

The gradient descent update is

$$\theta_{k+1} = \theta_k - \eta \nabla L(\theta_k), \quad (2.1)$$

where $\eta > 0$ is the learning rate.

Learning Rate

The learning rate η controls the size of each step. It is not learned by the basic algorithm; it is chosen as a configuration parameter.

3 A One-Dimensional Example

Let $L(\theta) = \theta^2$. Then $L'(\theta) = 2\theta$, so the update becomes

$$\theta_{k+1} = \theta_k - 2\eta\theta_k = (1 - 2\eta)\theta_k. \quad (3.1)$$

Worked Example

If $\eta = 0.1$ and $\theta_0 = 10$, then $\theta_1 = 8$, $\theta_2 = 6.4$, and the iterates move toward the minimizer $\theta = 0$.

Warning

A learning rate that is too large can overshoot the minimum and may even make the iterates diverge.

Key Result

Gradient descent is local: each step uses slope information at the current point, not a global map of the whole loss surface.